Yaxin Luo

Post-Training · Reinforcement Learning · Long-Horizon Agents

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Education

Aug 2025-Present	**MBZUAI** · Ph.D. in Machine Learning	Advisor: Dr. Zhiqiang Shen
Sep 2021-Apr 2025	**Technical University of Denmark** · B.Sc. in General Engineering	Advisor: Prof. Dimitrios Papadopoulos
Sep 2020-Apr 2021	**University of Edinburgh** · B.Sc. in Mathematics	

Research Interest

I work on **post-training** and **reinforcement learning** for agents that do long, expert work. A harness can make a model carry out long-horizon tasks today; I move that ability into the weights, building the whole pipeline: harnesses that produce training-grade trajectories, RL environments, evaluation benchmarks and reward design, then mid-training, SFT and RL. My long-term goal is agents that carry out arbitrarily complex, long-horizon expert work in digital environments, with a path toward real-world tasks; my current testbeds are agentic design and game development.

Experience

Meituan · M17 LongCat — Research Intern, Northern Dipper Talent Program — Sep 2026-Present

Built the harness base that **LongCat 2.5**'s Agentic Design post-training runs on, and lead its harness-optimization and harness-internalization research: evolve the harness on a task, then train what it teaches into the weights.

Evolvarium: the harness base for post-training

- Designed and built **Evolvarium** (open source), the base the team's post-training iterates on across **3D modeling, slides, web and poster design**, lifting LongCat 2.5's aesthetic and design ability substantially. Every run is sandboxed and recorded (inputs, harness revision, trace, grade) and exported as LongCat SFT data.
- **Meta-optimized harness (MoH)**: a Meta Agent edits instructions, skills, hooks and tools; a keep rule calibrated to measured noise keeps or reverts each edit. On our slides benchmark, the evolved harness lifts Kimi K3 from **57.6 to 74.7** (design +9.2, motion +9.5) and Qwen3.8-27B from **52.6 to 65.0**, with no weight change.
- Showed the grader is what the optimizer learns: a stricter grader cut every gain by a third to 70%; 8 of 9 harnesses kept after seeing failed rubric items copied it (0 of 11 with outcome-only feedback); write-up and 12 lineages.

Internalizing the harness: SFT, then RL

- Goal: a student model in a plain harness (Claude Code, Pi) matching a teacher in the evolved one, with the harness's preferences, design principles and reusable scripts in its weights. Compared SFT recipes from a 20k synthesized slides pool: naive distillation, evolved-harness trajectories, offline turn-by-turn **rewriting**, online **teacher intervention** on student turns (to internalize tools and middleware), and **runbook + hint** distillation with leak audits.
- Co-designed the **tiered RL reward**: validity, pixel-level usability, content against frozen per-query checklists, then a blind agentic aesthetics judge. Tier first, taste within tier, so a broken deck never outranks a usable one; the judge orders 59 of 60 calibration pairs. Fixed a GRPO zero-advantage failure by scoring broken tiers on bad pages.

Now and next

- Running SFT on Qwen3.8-27B for each recipe, then RL with the tiered reward. Next, building the full post-training pipeline for **game development**: the harness, RL environments, evaluation benchmarks and RL reward design.

Meituan · M17 LongCat — Research Intern — Apr 2026-Sep 2026

- Led **AutoDesign** to explore long-horizon design agents and harness methods, validating an end-to-end loop of reusable DesignHarnesses, Designer-Critic iteration, evaluation protocols and real deliverables (posters, slides, webpages and videos from a single paper), the systems foundation for the post-training work above.

MBZUAI · VILA Lab — Research Assistant — Jan-Aug 2025

- Studied how language-pretrained parameters transfer to vision-only tasks through label-free random-label bridge training; built Open CaptchaWorld, an interactive benchmark for multimodal browser agents (NeurIPS 2025).

Open-Source GitHub Projects

AutoDesign 238 Stars · 2026.10.09 **2026-Present**

An open-source, model-agnostic Design Agent that turns a paper into an editable poster, slide deck, webpage and subtitled video on Codex, Claude Code or Kimi Code. Its **Meta-Harness Optimization** improves reusable DesignHarnesses from traces and evaluators behind acceptance gates; on the 100-paper PosterBench it scores **78.32**, **7.45** points above Claude Design, and gains **+5.01 to +19.56** across seven matched configurations.

github.com/Yaxin9Luo/AutoDesign

FigMirror 523 Stars · 2026.10.09 **2026-Present**

A reference-image-driven scientific figure agent (Orchestrator-Drawer-Reviewer with Grounded Measurement) that turns user data into editable Matplotlib code and camera-ready PDFs; I wrote the early method, Web UI and refinement specifications and the Codex / Claude runners and integration tests.

github.com/VILA-Lab/FigMirror

Publications * EQUAL CONTRIBUTION · FULL LIST ON GOOGLE SCHOLAR

1. **ICML 2026** · Next-Gen CAPTCHAs: Leveraging the Cognitive Gap for Scalable and Diverse GUI-Agent Defense · co-first author
2. **arXiv 2026** · AutoDesign: Meta-Harness Optimization for Long-Horizon Agentic Design · first author
3. **ACL 2026 Main** · LLMSurgeon: Diagnosing Data Mixture of Large Language Models · first author
4. **NeurIPS 2025** · Open CaptchaWorld: A Comprehensive Web-based Platform for Testing and Benchmarking Multimodal LLM Agents · first author
5. **ICLR 2025** · γ-MoD: Exploring Mixture-of-Depth Adaptation for Multimodal Large Language Models · first author
6. **ECCV 2024** · APL: Anchor-based Prompt Learning for One-stage Weakly Supervised Referring Expression Comprehension · first author

Additional Experience

- IEEE Cybermatics Congress 2024: volunteer organizing-team member; served as conference assistant and session chair for the Smart Data workshop.
- SciSec 2024: conference volunteer. Institute of Social Science Survey, Peking University: Summer Intern, Public Mental Health Project (Jul-Sep 2019).